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26 July 1976
SUPERSEDING
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13 March 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, BACKWARD-WAVE OSCILLATOR

TYPE 7824

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Type 0, 5.4 to 5.9 GHz, 50 milliwatts, solenoid focused, air cooled.

ABSOLUTE RATINGS:

Parameter:	Ef	Ec	Ew	Eb	Ik	Ic	Iw	Load Z
Unit:	V	Vdc	Vdc	Vdc	mAdc	mAdc	mAdc	ohms
Maximum:	6.3	250	900	200	10	1	3	---
Minimum:	---	---	---	0	---	---	---	50

ABSOLUTE RATINGS:

Parameter:	Ambient Temperature	Tube Seal Temperature	tk	Magnetic field
Unit:	°C	°C	sec	Gauss
Maximum:	80	200	90	1000
Minimum:	-65	---	---	750

PHYSICAL CHARACTERISTICS:

Mounting:	Any
Dimensions:	See outline drawing
Weight:	1 pound approximate

TEST CONDITIONS:

Parameter:	Ef	Ec	Ew	Eb	Ik	Ambient Temperature	Magnetic Field
Unit:	V	Vdc	Vdc	Vdc	mAdc	°C	Gauss
	6.3	See note 1	See note 1	150	See note 1	15	800

REQUIREMENT OR TEST:

First Article Test - Required (See notes 8 and 9)

7824
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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	First Article Test	See Note 8				
1042	Shock	15 G; 11 ms; MIL-STD-202; Method 205C	---	---	---	---
1011	Humidity	Relative humidity 95 to 100% at 95 to 100°C 96 hours	---	---	---	---
	Quality conformance inspection, part 1					
1256	Anode current	F = 5.4 thru 5.9 GHz	Ic	-0.1	0.5	mAdc
1256	Helix current	F = 5.9 GHz	Iw	---	3.0	mAdc
1256	Cathode current	F = 5.9 GHz	Ik	---	10.0	mAdc
1256	Collector current	F = 5.9 GHz	Ib	---	8	mAdc
1261	Helix voltage (1)	F = 5.4 GHz	Ew	525	---	Vdc
1261	Helix voltage (2)	F = 5.9 GHz	Ew	---	825	Vdc
1261	Helix voltage variation (3)	F = 5.4 thru 5.9 GHz See notes 4 and 5	ΔEw	125	225	Vdc
1261	Anode voltage	F = 5.4 GHz	Ec	50	210	Vdc
----	Frequency stability	F = 5.4 thru 5.9 GHz See note 7	ΔF	---	1000	Hz
4281	Heater current	E_f = 6.3 V	If	0.5	1.0	A
----	Insulation	Ef = 0; See note 3	---	---	---	---
4250	Power output levels	F = 5.4 thru 5.9 GHz	Po	50	---	mW
4251	Power output variation	F = 5.4 thru 5.9 GHz See Note 6	ΔPo	----	3	db
	Quality conformance inspection, part 2					
D-30(b)	Dimensions	See Figure 1	---	---	---	---
	Quality conformance inspection, part 3					
----	Life test	Group D; F = 5.4 thru 5.9 GHz Swept frequency operation	t	500	---	hrs
	Life test end points:					
4250	Power output level	F = 5.4 thru 5.9 GHz	Po	40	---	mW
1261	Anode voltage	F = 5.4 thru 5.9 GHz	Ec	50	210	Vdc
1256	Anode current	F = 5.4 thru 5.9 GHz	Ic	-0.1	0.5	mAdc
1256	Cathode current	F = 5.4 thru 5.9 GHz	Ik	---	10	mAdc

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NOTES:

1. With the helix voltage Ew set to the value supplied for 5.4 GHz operation on data sheet adjust the anode voltage (Ec) for specified value of Ik (value shall be supplied with each tube and shall not be greater than 10 mAdc).

2. Suitable heat sink to limit temperature rise of tube to be provided by associated solenoid end plates through tube to solenoid mounting clamps. Solenoid temperature not to exceed 30°C.

3. Test potentials shall be applied one at a time for 30 seconds each.

Test potential with respect to ground	Points connected to test potential	Maximum Leakage limit
Vdc	(all others grounded)	µAdc
-4000	Pin B	25
-4000	Pin A	25
-4000	Pins B, D, E	25

4. The curve of frequency versus the logarithm of helix voltage of an individual tube shall not deviate from a straight line by more than plus or minus 0.4 percent in frequency.

5. ΔEw is defined as the helix voltage spread of an individual tube to cover the frequency range of 5.4 to 5.9 GHz.

6. ΔPo is defined as the difference between the highest and lowest Po within the band.

7. ΔF is defined as the peak-to-peak frequency deviation (short term stability) about the operating frequency (F) as measured with the LFE-5024C Microwave Stalo Tester or equivalent. An equivalent method would be to utilize a calibrated discriminator that reads instantaneous microwave frequency change, and samples that frequency change over a 2 ms. time interval.

8. These tests are to be performed only when required by the contract or when quality deficiency evaluation warrants verification of test results.

9. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this specification sheet. In addition, first article sample approval shall include satisfactory demonstration of tube-to-system compatibility. Invitation for bids should provide that the preparing activity reserves right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Custodian:
Air Force - 85

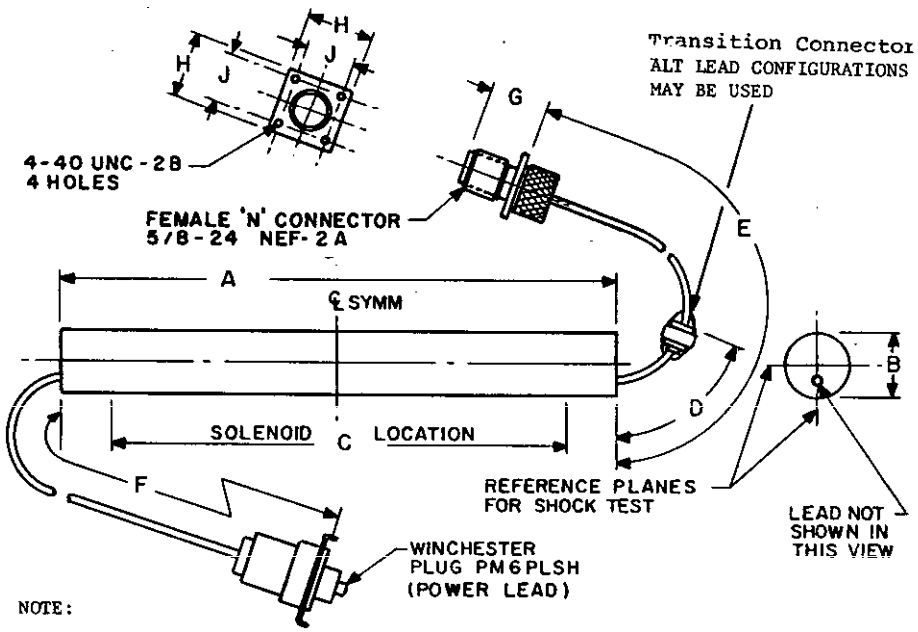
Preparing activity:
Air Force - 85

Review activity:
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User activity:
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NOTE:

Metric equivalents (to the nearest .01 mm) are given for general information only and based upon 1 inch = 25.4 mm.

POWER LEAD CONNECTORS	
PIN	CONNECTIONS
A	COLLECTOR
B	ANODE
C	HELIX
D	CATHODE
E	HEATER
F	GROUND

Quality conformance inspection, part 2 (all dimensions)

SYM	INCHES			MILLIMETERS			SYM	INCHES			MILLIMETERS		
	MIN	NOM	MAX	MIN	NOM	MAX		MIN	NOM	MAX	MIN	NOM	MAX
A	10.875	10.937	10.999	276.23	277.80	279.37	F	11.500	12.000	12.500	292.10	304.80	317.50
B	.969	1.000	1.031	24.61	25.40	26.19	G		.870			22.10	
C	10.000		10.031	254.00		254.79	H	.985	1.000	1.015	25.02	25.40	25.78
D			1.625			41.28	J	.712	.718	.722	18.08	18.24	18.34
E	18.000	18.500	19.000	457.20	469.90	482.60							

FIGURE 1 Outline drawing for tube type 7824

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